

OVP Guide to Using Processor Models

Model specific information for ARM_Cortex-R8MPx3

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Contents

1	Overview	1
1.1	Description	1
1.2	Licensing	1
1.3	Limitations	2
1.4	Verification	2
1.5	Features	3
1.5.1	Core Features	3
1.5.2	Memory System	3
1.5.3	Advanced SIMD and Floating-Point Features	3
1.5.4	Generic Interrupt Controller	3
1.6	Debug Mask	4
1.7	AArch32 Unpredictable Behavior	4
1.7.1	Equal Target Registers	4
1.7.2	Floating Point Load/Store Multiple Lists	4
1.7.3	Floating Point VLD[2-4]/VST[2-4] Range Overflow	5
1.7.4	If-Then (IT) Block Constraints	5
1.7.5	Use of R13	5
1.7.6	Use of R15	5
1.7.7	Unpredictable Instructions in Some Modes	5
1.8	Integration Support	6
1.8.1	Halt Reason Introspection	6
1.8.2	Automatic WFI Wake Up	6
1.8.3	System Register Access Monitor	6
1.8.4	System Register Implementation	7
2	Configuration	8
2.1	Location	8
2.2	GDB Path	8
2.3	Semi-Host Library	8
2.4	Processor Endian-ness	8
2.5	QuantumLeap Support	8
2.6	Processor ELF code	8
3	All Variants in this model	9
4	Bus Master Ports	13
5	Bus Slave Ports	14

6	Net Ports	15
7	FIFO Ports	20
8	Formal Parameters	21
8.1	Parameter values and limits	27
8.2	Parameter values and limits	31
9	Execution Modes	33
10	Exceptions	34
11	Hierarchy of the model	35
11.1	Level 1: MPCORE	35
11.2	Level 2: CPU	35
12	Model Commands	37
12.1	Level 1: MPCORE	37
12.1.1	isync	37
12.1.2	itrace	37
12.2	Level 2: CPU	38
12.2.1	debugflags	38
12.2.2	isync	38
12.2.3	itrace	38
12.2.4	listSysRegsAA32	39
12.2.4.1	Argument description	39
13	Registers	40
13.1	Level 1: MPCORE	40
13.2	Level 2: CPU	40
13.2.1	Core	40
13.2.2	Control	40
13.2.3	User	41
13.2.4	FIQ	41
13.2.5	IRQ	41
13.2.6	Supervisor	41
13.2.7	Undefined	42
13.2.8	Abort	42
13.2.9	SIMD_VFP	42
13.2.10	SIMD_VFP_SYS	42
13.2.11	Coprocessor_32_bit	43
13.2.12	Integration_support	44
13.2.13	MPCore_SCR	44
13.2.14	MPCore_distributor	45
13.2.15	MPCore_processor_interface	47
13.2.16	MPCore_timer_and_watchdog	47
13.2.17	MPCore_global_timer	47

Chapter 1

Overview

This document provides the details of an OVP Fast Processor Model variant.

OVP Fast Processor Models are written in C and provide a C API for use in C based platforms. The models also provide a native interface for use in SystemC TLM2 platforms.

The models are written using the OVP VMI API that provides a Virtual Machine Interface that defines the behavior of the processor. The VMI API makes a clear line between model and simulator allowing very good optimization and world class high speed performance. Most models are provided as a binary shared object and also as source. This allows the download and use of the model binary or the use of the source to explore and modify the model.

The models are run through an extensive QA and regression testing process and most model families are validated using technology provided by the processor IP owners. There is a companion document (OVP Guide to Using Processor Models) which explains the general concepts of OVP Fast Processor Models and their use. It is downloadable from the OVPworld website documentation pages.

1.1 Description

ARM Processor Model

1.2 Licensing

Usage of binary model under license governing simulator usage.

Note that for models of ARM CPUs the license includes the following terms:

Licensee is granted a non-exclusive, worldwide, non-transferable, revocable licence to:

If no source is being provided to the Licensee: use and copy only (no modifications rights are granted) the model for the sole purpose of designing, developing, analyzing, debugging, testing, verifying, validating and optimizing software which: (a) (i) is for ARM based systems; and (ii) does not incorporate the ARM Models or any part thereof; and (b) such ARM Models may not be used

to emulate an ARM based system to run application software in a production or live environment.

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In the case of any Licensee who is either or both an academic or educational institution the purposes shall be limited to internal use.

Except to the extent that such activity is permitted by applicable law, Licensee shall not reverse engineer, decompile, or disassemble this model. If this model was provided to Licensee in Europe, Licensee shall not reverse engineer, decompile or disassemble the Model for the purposes of error correction.

The License agreement does not entitle Licensee to manufacture in silicon any product based on this model.

The License agreement does not entitle Licensee to use this model for evaluating the validity of any ARM patent.

Source of model available under separate Imperas Software License Agreement.

1.3 Limitations

Instruction pipelines are not modeled in any way. All instructions are assumed to complete immediately. This means that instruction barrier instructions (e.g. ISB, CP15ISB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. The model does not implement speculative fetch behavior. The branch cache is not modeled.

Caches and write buffers are not modeled in any way. All loads, fetches and stores complete immediately and in order, and are fully synchronous (as if the memory was of Strongly Ordered or Device-nGnRnE type). Data barrier instructions (e.g. DSB, CP15DSB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. Cache manipulation instructions are implemented as NOPs, with the exception of any undefined instruction behavior, which is modeled.

Real-world timing effects are not modeled: all instructions are assumed to complete in a single cycle.

Performance Monitors are implemented as a register interface only except for the cycle counter, which is implemented assuming one instruction per cycle.

1.4 Verification

Models have been extensively tested by Imperas. ARM Cortex models have been successfully used by customers to simulate SMP Linux, Ubuntu Desktop, VxWorks and ThreadX on Xilinx Zynq virtual platforms.

1.5 Features

The precise set of implemented features in the model is defined by ID registers. Use overrides to modify these if required (for example `override_PFR0` or `override_AA64PFR0_EL1`).

1.5.1 Core Features

Thumb-2 instructions are supported.

Trivial Jazelle extension is implemented.

1.5.2 Memory System

FCSE extension is implemented.

PMSA address translation is implemented with a unified MPU with 12 regions. Use parameter “`override_MPUIR`” to specify a different MPU size if required.

1 ITCM is implemented.

1 DTCM is implemented.

For each TCM, a corresponding bus port is present. If parameter “`useInternalTCMs`” is `True`, TCMs are modelled internally and the port is a target port, allowing TCM contents to be examined and updated externally. Otherwise, if parameter “`useInternalTCMs`” is `False`, the corresponding port is an initiator port to which an external TCM memory model must be connected.

1.5.3 Advanced SIMD and Floating-Point Features

VFP instructions are implemented.

The model implements trapped exceptions if `FPTrap` is set to 1 in `MVFR0` (for AArch32) or `MVFR0_EL1` (for AArch64). When floating point exception traps are taken, cumulative exception flags are not updated (in other words, cumulative flag state is always the same as prior to instruction execution, even for SIMD instructions). When multiple enabled exceptions are raised by a single floating point operation, the exception reported is the one in least-significant bit position in `FPSCR` (for AArch32) or `FPCR` (for AArch64). When multiple enabled exceptions are raised by different SIMD element computations, the exception reported is selected from the lowest-index-number SIMD operation. Contact Imperas if requirements for exception reporting differ from these.

Trapped exceptions not are implemented in this variant (`FPTrap=0`)

1.5.4 Generic Interrupt Controller

GIC block is implemented (GICv1). Accesses to GIC registers can be viewed externally by connecting to the 32-bit `GICRegisters` bus port. Secure register accesses are at offset 0x0 on this bus; for example, a secure access to GIC register `ICDDCR` can be observed by monitoring address

0x00001000. Non-secure accesses are at offset 0x80000000 on this bus; for example, a non-secure access to GIC register ICDDCR can be observed by monitoring address 0x80001000

1.6 Debug Mask

It is possible to enable model debug features in various categories. This can be done statically using the “override_debugMask” parameter, or dynamically using the “debugflags” command. Enabled debug features are specified using a bitmask value, as follows:

Value 0x004: enable debugging of MMU/MPU mappings.

Value 0x020: enable debugging of reads and writes of GIC block registers.

Value 0x040: enable debugging of exception routing via the GIC model component.

Value 0x080: enable debugging of all system register accesses.

Value 0x100: enable debugging of all traps of system register accesses.

Value 0x200: enable verbose debugging of other miscellaneous behavior (for example, the reason why a particular instruction is undefined).

Value 0x400: enable debugging of Performance Monitor timers

All other bits in the debug bitmask are reserved and must not be set to non-zero values.

1.7 AArch32 Unpredictable Behavior

Many AArch32 instruction behaviors are described in the ARM ARM as CONSTRAINED UNPREDICTABLE. This section describes how such situations are handled by this model.

1.7.1 Equal Target Registers

Some instructions allow the specification of two target registers (for example, double-width SMULL, or some VMOV variants), and such instructions are CONSTRAINED UNPREDICTABLE if the same target register is specified in both positions. In this model, such instructions are treated as UNDEFINED.

1.7.2 Floating Point Load/Store Multiple Lists

Instructions that load or store a list of floating point registers (e.g. VSTM, VLDM, VPUSH, VPOP) are CONSTRAINED UNPREDICTABLE if either the uppermost register in the specified range is greater than 32 or (for 64-bit registers) if more than 16 registers are specified. In this model, such instructions are treated as UNDEFINED.

1.7.3 Floating Point VLD[2-4]/VST[2-4] Range Overflow

Instructions that load or store a fixed number of floating point registers (e.g. VST2, VLD2) are CONSTRAINED UNPREDICTABLE if the upper register bound exceeds the number of implemented floating point registers. In this model, these instructions load and store using modulo 32 indexing (consistent with AArch64 instructions with similar behavior).

1.7.4 If-Then (IT) Block Constraints

Where the behavior of an instruction in an if-then (IT) block is described as CONSTRAINED UNPREDICTABLE, this model treats that instruction as UNDEFINED.

1.7.5 Use of R13

In architecture variants before ARMv8, use of R13 was described as CONSTRAINED UNPREDICTABLE in many circumstances. From ARMv8, most of these situations are no longer considered unpredictable. This model allows R13 to be used like any other GPR, consistent with the ARMv8 specification.

1.7.6 Use of R15

Use of R15 is described as CONSTRAINED UNPREDICTABLE in many circumstances. This model allows such use to be configured using the parameter “unpredictableR15” as follows:

Value “undefined”: any reference to R15 in such a situation is treated as UNDEFINED;

Value “nop”: any reference to R15 in such a situation causes the instruction to be treated as a NOP;

Value “raz_wi”: any reference to R15 in such a situation causes the instruction to be treated as a RAZ/WI (that is, R15 is read as zero and write-ignored);

Value “execute”: any reference to R15 in such a situation is executed using the current value of R15 on read, and writes to R15 are allowed (but are not interworking).

Value “assert”: any reference to R15 in such a situation causes the simulation to halt with an assertion message (allowing any such unpredictable uses to be easily identified).

In this variant, the default value of “unpredictableR15” is “undefined”.

1.7.7 Unpredictable Instructions in Some Modes

Some instructions are described as CONSTRAINED UNPREDICTABLE in some modes only (for example, MSR accessing SPSR is CONSTRAINED UNPREDICTABLE in User and System modes). This model allows such use to be configured using the parameter “unpredictableModal”, which can have values “undefined” or “nop”. See the previous section for more information about the meaning of these values.

In this variant, the default value of “unpredictableModal” is “nop”.

1.8 Integration Support

This model implements a number of non-architectural pseudo-registers and other features to facilitate integration.

1.8.1 Halt Reason Introspection

An artifact register HaltReason can be read to determine the reason or reasons that a processor is halted. This register is a bitfield, with the following encoding: bit 0 indicates the processor has executed a wait-for-event (WFE) instruction; bit 1 indicates the processor has executed a wait-for-interrupt (WFI) instruction; and bit 2 indicates the processor is held in reset.

1.8.2 Automatic WFI Wake Up

A simulation environment can cause a core to wake up from the WFI state by writing to the artifact register WakeWFINow.

1.8.3 System Register Access Monitor

If parameter “enableSystemMonitorBus” is True, an artifact 32-bit bus “SystemMonitor” is enabled for each PE. Every system register read or write by that PE is then visible as a read or write on this artifact bus, and can therefore be monitored using callbacks installed in the client environment (use opBusReadMonitorAdd/opBusWriteMonitorAdd or icmAddBusReadCallback-/icmAddBusWriteCallback, depending on the client API). The format of the address on the bus is as follows:

bits 31:26 - zero

bit 25 - 1 if AArch64 access, 0 if AArch32 access

bit 24 - 1 if non-secure access, 0 if secure access

bits 23:20 - CRm value

bits 19:16 - CRn value

bits 15:12 - op2 value

bits 11:8 - op1 value

bits 7:4 - op0 value (AArch64) or coprocessor number (AArch32)

bits 3:0 - zero

As an example, to view non-secure writes to writes to CNTFRQ_EL0 in AArch64 state, install a write monitor on address range 0x020e0330:0x020e0333.

1.8.4 System Register Implementation

If parameter “enableSystemBus” is True, an artifact 32-bit bus “System” is enabled for each PE. Slave callbacks installed on this bus can be used to implement modified system register behavior (use opBusSlaveNew or icmMapExternalMemory, depending on the client API). The format of the address on the bus is the same as for the system monitor bus, described above.

Chapter 2

Configuration

2.1 Location

This model's VLVN is `arm.ovpworld.org/processor/arm/1.0`.

The model source is usually at:

`$IMPERAS_HOME/ImperasLib/source/arm.ovpworld.org/processor/arm/1.0`

The model binary is usually at:

`$IMPERAS_HOME/lib/$IMPERAS_ARCH/ImperasLib/arm.ovpworld.org/processor/arm/1.0`

2.2 GDB Path

The default GDB for this model is: `$IMPERAS_HOME/lib/$IMPERAS_ARCH/gdb/arm-none-eabi-gdb`.

2.3 Semi-Host Library

The default semi-host library file is `arm.ovpworld.org/semihosting/armNewlib/1.0`

2.4 Processor Endian-ness

This is a LITTLE endian model.

2.5 QuantumLeap Support

This processor is qualified to run in a QuantumLeap enabled simulator.

2.6 Processor ELF code

The ELF code supported by this model is: `0x28`.

Chapter 3

All Variants in this model

This model has these variants

Variant	Description
ARMv4T	
ARMv4xM	
ARMv4	
ARMv4TxM	
ARMv5xM	
ARMv5	
ARMv5TxM	
ARMv5T	
ARMv5TExP	
ARMv5TE	
ARMv5TEJ	
ARMv6	
ARMv6K	
ARMv6T2	
ARMv6KZ	
ARMv7	
ARM7TDMI	
ARM7EJ-S	
ARM720T	
ARM920T	
ARM922T	
ARM926EJ-S	
ARM940T	
ARM946E	
ARM966E	
ARM968E-S	
ARM1020E	
ARM1022E	
ARM1026EJ-S	
ARM1136J-S	
ARM1156T2-S	

ARM1176JZ-S	
Cortex-R4	
Cortex-R4F	
Cortex-R5MPx1	
Cortex-R5MPx2	
Cortex-R8MPx1	
Cortex-R8MPx2	
Cortex-R8MPx3	(described in this document)
Cortex-R8MPx4	
Cortex-R52MPx1	
Cortex-R52MPx2	
Cortex-R52MPx3	
Cortex-R52MPx4	
Cortex-R52+MPx1	
Cortex-R52+MPx2	
Cortex-R52+MPx3	
Cortex-R52+MPx4	
Cortex-R82MPx1	
Cortex-R82MPx2	
Cortex-R82MPx3	
Cortex-R82MPx4	
Cortex-R82MPx5	
Cortex-R82MPx6	
Cortex-R82MPx7	
Cortex-R82MPx8	
Cortex-A5UP	
Cortex-A5MPx1	
Cortex-A5MPx2	
Cortex-A5MPx3	
Cortex-A5MPx4	
Cortex-A8	
Cortex-A9UP	
Cortex-A9MPx1	
Cortex-A9MPx2	
Cortex-A9MPx3	
Cortex-A9MPx4	
Cortex-A7UP	
Cortex-A7MPx1	
Cortex-A7MPx2	
Cortex-A7MPx3	
Cortex-A7MPx4	
Cortex-A15UP	
Cortex-A15MPx1	
Cortex-A15MPx2	
Cortex-A15MPx3	

Cortex-A15MPx4	
Cortex-A17MPx1	
Cortex-A17MPx2	
Cortex-A17MPx3	
Cortex-A17MPx4	
AArch32	
AArch64	
Cortex-A32MPx1	
Cortex-A32MPx2	
Cortex-A32MPx3	
Cortex-A32MPx4	
Cortex-A35MPx1	
Cortex-A35MPx2	
Cortex-A35MPx3	
Cortex-A35MPx4	
Cortex-A53MPx1	
Cortex-A53MPx2	
Cortex-A53MPx3	
Cortex-A53MPx4	
Cortex-A55MPx1	
Cortex-A55MPx2	
Cortex-A55MPx3	
Cortex-A55MPx4	
Cortex-A57MPx1	
Cortex-A57MPx2	
Cortex-A57MPx3	
Cortex-A57MPx4	
Cortex-A65AEMPx1	
Cortex-A65AEMPx2	
Cortex-A65AEMPx3	
Cortex-A65AEMPx4	
Cortex-A72MPx1	
Cortex-A72MPx2	
Cortex-A72MPx3	
Cortex-A72MPx4	
Cortex-A73MPx1	
Cortex-A73MPx2	
Cortex-A73MPx3	
Cortex-A73MPx4	
Cortex-A75MPx1	
Cortex-A75MPx2	
Cortex-A75MPx3	
Cortex-A75MPx4	
Cortex-A76MPx1	
Cortex-A76MPx2	

Cortex-A76MPx3	
Cortex-A76MPx4	
Cortex-A78MPx1	
Cortex-A78MPx2	
Cortex-A78MPx3	
Cortex-A78MPx4	
MultiCluster	

Table 3.1: All Variants in this model

Chapter 4

Bus Master Ports

This model has these bus master ports.

Name	min	max	Connect?	Description
LLPP	32	32	optional	Low-latency peripheral port
ITCM0_CPU0	13	13	optional	instruction TCM
DTCM0_CPU0	13	13	optional	data TCM
FPP_CPU0	32	32	optional	Fast peripheral port
ITCM0_CPU1	13	13	optional	instruction TCM
DTCM0_CPU1	13	13	optional	data TCM
FPP_CPU1	32	32	optional	Fast peripheral port
ITCM0_CPU2	13	13	optional	instruction TCM
DTCM0_CPU2	13	13	optional	data TCM
FPP_CPU2	32	32	optional	Fast peripheral port
INSTRUCTION	32	32	mandatory	
DATA	32	32	optional	
GICRegisters	32	32	optional	GIC memory-mapped register block

Table 4.1: Bus Master Ports

Chapter 5

Bus Slave Ports

This model has no bus slave ports.

Chapter 6

Net Ports

This model has these net ports.

Name	Type	Connect?	Description
SPI32	input	optional	Shared peripheral interrupt
SPI33	input	optional	Shared peripheral interrupt
SPI34	input	optional	Shared peripheral interrupt
SPI35	input	optional	Shared peripheral interrupt
SPI36	input	optional	Shared peripheral interrupt
SPI37	input	optional	Shared peripheral interrupt
SPI38	input	optional	Shared peripheral interrupt
SPI39	input	optional	Shared peripheral interrupt
SPI40	input	optional	Shared peripheral interrupt
SPI41	input	optional	Shared peripheral interrupt
SPI42	input	optional	Shared peripheral interrupt
SPI43	input	optional	Shared peripheral interrupt
SPI44	input	optional	Shared peripheral interrupt
SPI45	input	optional	Shared peripheral interrupt
SPI46	input	optional	Shared peripheral interrupt
SPI47	input	optional	Shared peripheral interrupt
SPI48	input	optional	Shared peripheral interrupt
SPI49	input	optional	Shared peripheral interrupt
SPI50	input	optional	Shared peripheral interrupt
SPI51	input	optional	Shared peripheral interrupt
SPI52	input	optional	Shared peripheral interrupt
SPI53	input	optional	Shared peripheral interrupt
SPI54	input	optional	Shared peripheral interrupt
SPI55	input	optional	Shared peripheral interrupt
SPI56	input	optional	Shared peripheral interrupt
SPI57	input	optional	Shared peripheral interrupt
SPI58	input	optional	Shared peripheral interrupt
SPI59	input	optional	Shared peripheral interrupt
SPI60	input	optional	Shared peripheral interrupt
SPI61	input	optional	Shared peripheral interrupt

SPI62	input	optional	Shared peripheral interrupt
SPI63	input	optional	Shared peripheral interrupt
SPI64	input	optional	Shared peripheral interrupt
SPI65	input	optional	Shared peripheral interrupt
SPI66	input	optional	Shared peripheral interrupt
SPI67	input	optional	Shared peripheral interrupt
SPI68	input	optional	Shared peripheral interrupt
SPI69	input	optional	Shared peripheral interrupt
SPI70	input	optional	Shared peripheral interrupt
SPI71	input	optional	Shared peripheral interrupt
SPI72	input	optional	Shared peripheral interrupt
SPI73	input	optional	Shared peripheral interrupt
SPI74	input	optional	Shared peripheral interrupt
SPI75	input	optional	Shared peripheral interrupt
SPI76	input	optional	Shared peripheral interrupt
SPI77	input	optional	Shared peripheral interrupt
SPI78	input	optional	Shared peripheral interrupt
SPI79	input	optional	Shared peripheral interrupt
SPI80	input	optional	Shared peripheral interrupt
SPI81	input	optional	Shared peripheral interrupt
SPI82	input	optional	Shared peripheral interrupt
SPI83	input	optional	Shared peripheral interrupt
SPI84	input	optional	Shared peripheral interrupt
SPI85	input	optional	Shared peripheral interrupt
SPI86	input	optional	Shared peripheral interrupt
SPI87	input	optional	Shared peripheral interrupt
SPI88	input	optional	Shared peripheral interrupt
SPI89	input	optional	Shared peripheral interrupt
SPI90	input	optional	Shared peripheral interrupt
SPI91	input	optional	Shared peripheral interrupt
SPI92	input	optional	Shared peripheral interrupt
SPI93	input	optional	Shared peripheral interrupt
SPI94	input	optional	Shared peripheral interrupt
SPI95	input	optional	Shared peripheral interrupt
SPIVector	input	optional	Shared peripheral interrupt vectorized input
periphReset	input	optional	Peripheral reset (active high)
EVENTI	input	optional	Event input signal, active on rising edge
EVENTO	output	optional	Event output signal, active on rising edge
wdResetReq_CPU0	output	optional	Watchdog interrupt request
wdReset_CPU0	input	optional	Watchdog reset (active high)
scuReset	input	optional	SCU reset (active high)
MFILTEREN	input	optional	Configure Address Filtering enable bit (active high)
MFILTERSTART	input	optional	Configure AF start address for AXIM1

MFILTEREND	input	optional	Configure AF end address for AXIM1
PFILTERSTART	input	optional	Configure AF start address for AXIM1
PFILTEREND	input	optional	Configure AF end address for AXIM1
FPPSTART0	input	optional	Configure AF start address for FPP on core 0
FPPEND0	input	optional	Configure AF end address for FPP on core 0
FPPSTART1	input	optional	Configure AF start address for FPP on core 1
FPPEND1	input	optional	Configure AF end address for FPP on core 1
FPPSTART2	input	optional	Configure AF start address for FPP on core 2
FPPEND2	input	optional	Configure AF end address for FPP on core 2
FPPSTART3	input	optional	Configure AF start address for FPP on core 3
FPPEND3	input	optional	Configure AF end address for FPP on core 3
IRQOUT_CPU0	output	optional	IRQ wakeup
FIQOUT_CPU0	output	optional	FIQ wakeup
CLUSTERIDAFF1	input	optional	Configure MPIDR.Aff1
CLUSTERIDAFF2	input	optional	Configure MPIDR.Aff2
PERIPHBASE	input	optional	Base address for timers, watchdogs, interrupt controller, and SCU registers
VINITH1_CPU0	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND_CPU0	input	optional	Configure exception endianness (SCTLR.EE)
INITRAM1_CPU0	input	optional	Configure ITCM enable bit (active high)
TEINIT_CPU0	input	optional	Configure exception state at reset (SCTLR.TE)
CFGNMFI_CPU0	input	optional	Configure non-maskable fast interrupts (SCTLR.NMFI)
reset_CPU0	input	optional	Processor reset, active high
fiq_CPU0	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq_CPU0	input	optional	IRQ interrupt, active high (negation of nIRQ)
sei_CPU0	input	optional	System error interrupt, active on rising edge (negation of nSEI)
cpuHalt_CPU0	input	optional	Halt on reset
haltReason_CPU0	output	optional	Indicates why core is halted
AXI_SLVERR_CPU0	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
syncCluster_CPU0	output	optional	Cluster synchronization required
updateTimer_CPU0	output	optional	Internal timer update

FPUFLAGS_CPU0	output	optional	Cumulative exception flags signal output
SMPnAMP_CPU0	output	optional	SMP or AMP mode
wdResetReq_CPU1	output	optional	Watchdog interrupt request
wdReset_CPU1	input	optional	Watchdog reset (active high)
IRQOUT_CPU1	output	optional	IRQ wakeup
FIQOUT_CPU1	output	optional	FIQ wakeup
VINITHI_CPU1	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND_CPU1	input	optional	Configure exception endianness (SCTLR.EE)
INTRAML_CPU1	input	optional	Configure ITCM enable bit (active high)
TEINIT_CPU1	input	optional	Configure exception state at reset (SCTLR.TE)
CFGNMFI_CPU1	input	optional	Configure non-maskable fast interrupts (SCTLR.NMFI)
reset_CPU1	input	optional	Processor reset, active high
fiq_CPU1	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq_CPU1	input	optional	IRQ interrupt, active high (negation of nIRQ)
sei_CPU1	input	optional	System error interrupt, active on rising edge (negation of nSEI)
cpuHalt_CPU1	input	optional	Halt on reset
haltReason_CPU1	output	optional	Indicates why core is halted
AXI_SLVERR_CPU1	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
syncCluster_CPU1	output	optional	Cluster synchronization required
updateTimer_CPU1	output	optional	Internal timer update
FPUFLAGS_CPU1	output	optional	Cumulative exception flags signal output
SMPnAMP_CPU1	output	optional	SMP or AMP mode
wdResetReq_CPU2	output	optional	Watchdog interrupt request
wdReset_CPU2	input	optional	Watchdog reset (active high)
IRQOUT_CPU2	output	optional	IRQ wakeup
FIQOUT_CPU2	output	optional	FIQ wakeup
VINITHI_CPU2	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND_CPU2	input	optional	Configure exception endianness (SCTLR.EE)
INTRAML_CPU2	input	optional	Configure ITCM enable bit (active high)
TEINIT_CPU2	input	optional	Configure exception state at reset (SCTLR.TE)
CFGNMFI_CPU2	input	optional	Configure non-maskable fast interrupts (SCTLR.NMFI)
reset_CPU2	input	optional	Processor reset, active high
fiq_CPU2	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq_CPU2	input	optional	IRQ interrupt, active high (negation of nIRQ)

sei_CPU2	input	optional	System error interrupt, active on rising edge (negation of nSEI)
cpuHalt_CPU2	input	optional	Halt on reset
haltReason_CPU2	output	optional	Indicates why core is halted
AXI_SLVERR_CPU2	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
syncCluster_CPU2	output	optional	Cluster synchronization required
updateTimer_CPU2	output	optional	Internal timer update
FPUFLAGS_CPU2	output	optional	Cumulative exception flags signal output
SMPnAMP_CPU2	output	optional	SMP or AMP mode

Table 6.1: Net Ports

Chapter 7

FIFO Ports

This model has no FIFO ports.

Chapter 8

Formal Parameters

Name	Type	Description
verbose	Boolean	Specify verbosity of output
suppressCPSWarnings	Boolean	Suppress duplicate warnings generated using ARM_CP_CPSI or ARM_CP_CPSD message identifiers
showHiddenRegs	Boolean	Show hidden registers during register tracing
UAL	Boolean	Disassemble using UAL syntax
disableGICModel	Boolean	Disable the internal GIC model entirely
enableGICv2_64kB_Page	Boolean	Enable 64kB page size for GICv2 memory-mapped register groups (Xilinx Zynq Ultrascale support)
enableVFPAAtReset	Boolean	Enable vector floating point (SIMD and VFP) instructions at reset. (Enables cp10/11 in CPACR and sets FPEXC.EN)
useInternalTCMs	Boolean	Enable internally-modeled TCM memories
enableSystemBus	Boolean	Add 32-bit artifact System bus port, allowing system registers to be externally implemented
enableSystemMonitorBus	Boolean	Add 32-bit artifact SystemMonitor bus port, allowing system register accesses to be externally monitored
compatibility	Enumeration	Specify compatibility mode
	ISA	
	gdb	
	nopSVC	
unpredictableR15	Enumeration	Specify behavior for UNPREDICTABLE uses of AArch32 R15 register
	undefined	
	nop	
	raz_wi	
	execute	
	assert	

unpredictableModal	Enumeration	Specify behavior for UNPREDICTABLE instructions in certain AArch32 modes (for example, MRS using SPSR in System mode)
	undefined	
	nop	
	assert	
maxSIMDUnroll	Uns32	If SIMD operations are supported, specify the maximum number of parallel SIMD operations to unroll (unrolled operations can be faster, but produce more verbose JIT code)
override_debugMask	Uns32	Specifies debug mask, enabling debug output for model components
thumbNoCond	Boolean	Specify whether trapped Thumb instructions set CV=1 and COND field in syndrome (if False, both are zero)
enableHostAtomics	Boolean	Enable use of host atomic instructions to emulate load/store exclusive operations (not architecturally accurate, accelerates parallel simulation)
enableWFEHostAtomics	Boolean	Enable WFE and SEV instructions when using host atomics (does nothing otherwise)
override_numCPUs	Uns32	Specify the number of cores in a multiprocessor (maximum of 8 for GICv1/GICv2)
override_affinityMask	Uns32	Specify bitmask of implemented affinity bits in format Aff3:Aff2:Aff1:Aff0 (each a byte)
override_MPIDR_MT	Boolean	Specifies that processor is multi-threaded
override_MPIDR_Aff0	Uns32	Override Aff0 field in MPIDR/MPIDR_EL1 register
override_MPIDR_Aff1	Uns32	Override Aff1 field in MPIDR/MPIDR_EL1 register (also possible by writing CLUSTERIDAFF1 configuration net)
override_MPIDR_Aff2	Uns32	Override Aff2 field in MPIDR/MPIDR_EL1 register (also possible by writing CLUSTERIDAFF2 configuration net)
override_MPIDR_Aff3	Uns32	Override Aff3 field in MPIDR_EL1 register (also possible by writing CLUSTERIDAFF3 configuration net)
override_fcsePresent	Boolean	Specifies that FCSE is present (if true)
override_fpexcDexPresent	Boolean	Specifies that the FPEXC.DEX register field is implemented (if true)

override_advSIMDPresent	Boolean	Specifies that Advanced SIMD extensions are present (if true)
override_vfpPresent	Boolean	Specifies that VFP extensions are present (if true)
override_timerScaleFactor	Uns32	Specifies the fraction of MIPS rate to use for MPCore timers (generic timers or global/local/watchdogs depending on implementation). Defaults to 20 for generic timers, 2 for others
override_GICD_NSACRPresent	Boolean	Specifies that optional GICD_NSACR distributor registers are present (GICv2 only)
override_GICD_PPISRPresent	Boolean	Specifies that implementation-specific GICD_PPISR distributor register is present (GICv1 ICDP-PIS/ICPPISR, GICv1 and GICv2 only)
override_GICD_SPISRPresent	Boolean	Specifies that implementation-specific GICD_SPISR distributor registers are present (GICv1 ICD-SPIS/ICSPISR)
override_GIC_PPIMask	Uns32	Specify bitmask of implemented PPIs in the GIC (e.g. ID16 is 0x0001, ID31 is 0x8000)
override_GICCDISABLE	Boolean	Specify initial value of GICCDISABLE
allowMSRToCurrentBank	Boolean	Allow access to the current mode registers via the MSR/MRS (Banked Register) instructions
override_SCTLR_V	Boolean	Override SCTLR.V with the passed value (enables high vectors; also configurable using VINITHI pin)
override_SCTLR_IE	Boolean	Override SCTLR.IE with the passed value (configures instruction endianness; also configurable using CFGIE pin)
override_SCTLR_EE	Boolean	Override SCTLR.EE with the passed value (configures exception data endianness; also configurable using CFGEE pin)
override_SCTLR_TE	Boolean	Override SCTLR.TE with the passed value (configures Thumb state for exception handling; also configurable using TEINIT pin)
override_SCTLR_NMFI	Boolean	Override SCTLR.NMFI with the passed value (configures NMFI state for exception handling; also configurable using CFGNMFI pin)
override_SCTLR_CP15BEN_Present	Boolean	Enable ARMv7 SCTLR.CP15BEN bit (CP15 barrier enable)
override_MIDR	Uns32	Override MIDR/MIDR_EL1 register

override_CTR	Uns32	Override CTR/CTR_EL0 register
override_MPUIR	Uns32	Override MPUIR register
override_MPUIR_DRegion	Uns32	Override MPUIR.DRegion with the passed value
override_CLIDR	Uns32	Override CLIDR/CLIDR_EL1 register
override_ACTLR_SMP	Boolean	Override ACTLR SMP enable bit
override_TCMTR	Uns32	Override TCMTR register
override_AIDR	Uns32	Override AIDR/AIDR_EL1 register
override_ITCMRR_E	Boolean	Override ITCMRR register enable bit
override_ITCMRR_SIZE	Uns32	Override ITCMRR register size bits
override_tcmsPresent	Boolean	Specifies that TCM is present (if true)
override_DTCMRR_SIZE	Uns32	Override DTCMRR register size bits
override_CBAR	Uns32	Override Configuration Base Address Register (corresponds to value on PERIPHBASE configuration input)
override_PFR0	Uns32	Override ID_PFR0/ID_PFR0_EL1 register
override_PFR1	Uns32	Override ID_PFR1/ID_PFR1_EL1 register
override_DFR0	Uns32	Override ID_DFR0/ID_DFR0_EL1 register
override_AFR0	Uns32	Override ID_AFR0/ID_AFR0_EL1 register
override_MMFR0	Uns32	Override ID_MMFR0/ID_MMFR0_EL1 register
override_MMFR1	Uns32	Override ID_MMFR1/ID_MMFR1_EL1 register
override_MMFR2	Uns32	Override ID_MMFR2/ID_MMFR2_EL1 register
override_MMFR3	Uns32	Override ID_MMFR3/ID_MMFR3_EL1 register
override_ISAR0	Uns32	Override ID_ISAR0/ID_ISAR0_EL1 register
override_ISAR1	Uns32	Override ID_ISAR1/ID_ISAR1_EL1 register
override_ISAR2	Uns32	Override ID_ISAR2/ID_ISAR2_EL1 register

override_ISAR3	Uns32	Override ID_ISAR3/ID_ISAR3_EL1 register
override_ISAR4	Uns32	Override ID_ISAR4/ID_ISAR4_EL1 register
override_ISAR5	Uns32	Override ID_ISAR5/ID_ISAR5_EL1 register
override_PMCR	Uns32	Override PMCR/PMCR_EL0 register (not functionally significant in the model)
override_PMCEID0	Uns64	Override PMCEID0/PMCEID0_EL0 register (not functionally significant in the model)
override_PMCEID1	Uns64	Override PMCEID1/PMCEID1_EL0 register (not functionally significant in the model)
override_SACTLR	Uns32	Override Cortex-R profile SACTLR register (not functionally significant in the model)
override_BuildOptions0	Uns32	Override Cortex-R profile BuildOptions0 register (not functionally significant in the model)
override_BuildOptions1	Uns32	Override Cortex-R profile BuildOptions1 register (not functionally significant in the model)
override_DBGDIDR	Uns32	Override DBGDIDR register (not functionally significant in the model)
override_FPSID	Uns32	Override SIMD/VFP FPSID register
override_MVFR0	Uns32	Override SIMD/VFP MVFR0/MVFR0_EL1 register
override_MVFR1	Uns32	Override SIMD/VFP MVFR1/MVFR1_EL1 register
override_FPEXC	Uns32	Override SIMD/VFP FPEXC/FPEXC32_EL2 register
override_GICC.IIDR	Uns32	Override GICC.IIDR register (GICv1 ICCIHDR)
override_GICD.TYPER	Uns32	Override GICD.TYPER register (GICv1 ICDICTR)
override_GICD.TYPER.ITLines	Uns32	Override ITLinesNumber field of GICD.TYPER register (GICv1 ICDICTR)
override_GICD.ICFGRN	Uns32	Override reset value of GICD_ICFGR2...GICD_ICFGRn (GICv1 ICDICFR2...ICDICFRn)
override_GICD.IIDR	Uns32	Override GICD.IIDR register (GICv1 ICDIHDR)
override_GICH.VTR	Uns32	Override GICH.VTR register

override_ICCPMRBits	Uns32	Specify the number of writable bits in GICC.PMR (GICv1 IC-CPMR)
override_minICCBPR	Uns32	Specify the minimum possible value for GICC.BPR (GICv1 IC-CBPR)
override_ERG	Uns32	Specifies exclusive reservation granule
override_CCSIDR_1I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 1 instruction)
override_CCSIDR_1D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 1 data)
override_CCSIDR_2I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 2 instruction)
override_CCSIDR_2D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 2 data)
override_CCSIDR_3I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 3 instruction)
override_CCSIDR_3D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 3 data)
override_CCSIDR_4I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 4 instruction)
override_CCSIDR_4D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 4 data)
override_CCSIDR_5I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 5 instruction)
override_CCSIDR_5D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 5 data)
override_CCSIDR_6I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 6 instruction)
override_CCSIDR_6D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 6 data)
override_CCSIDR_7I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 7 instruction)
override_CCSIDR_7D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 7 data)
override_STRoffsetPC12	Boolean	Specifies that STR/STRB of PC should do so with 12:byte offset from the current instruction (if true), otherwise an 8:byte offset is used
override_mpuV5ExtAP	Boolean	Specifies that ARMv5 MPU extended access permissions implemented (cp15/5/2 and cp15/5/2)
override_ignoreBadCp15	Boolean	Specifies whether invalid coprocessor 15 access should be ignored (if true) or cause Invalid Instruction exceptions (if false)
override_SGIDisable	Boolean	Override whether GIC SGIs may be disabled (if true) or are permanently enabled (if false)
override_condUndefined	Boolean	Force undefined instructions to take Undefined Instruction exception even if they are conditional
override_deviceStrongAligned	Boolean	Force accesses to Device and Strongly Ordered regions to be aligned

override_Control_V	Boolean	Override SCTL.R.V with the passed value (deprecated, use override.SCTL.R.V)
override_MainId	Uns32	Override MIDR register (deprecated, use override.MIDR)
override_CacheType	Uns32	Override CTR register (deprecated, use override.CTR)
override_MPUType	Uns32	Override MPUIR register (deprecated, use override.MPUIR)
override_InstructionAttributes0	Uns32	Override ID_ISAR0 register (deprecated, use override.ISAR0)
override_InstructionAttributes1	Uns32	Override ID_ISAR1 register (deprecated, use override.ISAR1)
override_InstructionAttributes2	Uns32	Override ID_ISAR2 register (deprecated, use override.ISAR2)
override_InstructionAttributes3	Uns32	Override ID_ISAR3 register (deprecated, use override.ISAR3)
override_InstructionAttributes4	Uns32	Override ID_ISAR4 register (deprecated, use override.ISAR4)
override_InstructionAttributes5	Uns32	Override ID_ISAR5 register (deprecated, use override.ISAR5)
override_SCUControl_Address_filtering_enable	Boolean	Override SCU Control Address filtering enable bit in register
override_MasterFilteringStart	Uns32	Override MasterFilteringStart register
override_MasterFilteringEnd	Uns32	Override MasterFilteringEnd register
override_PeripheralsFilteringStart	Uns32	Override PeripheralsFilteringStart register
override_PeripheralsFilteringEnd	Uns32	Override PeripheralsFilteringEnd register
override_FPPFilteringStart0	Uns32	Override FPP0 Port filtering start address
override_FPPFilteringEnd0	Uns32	Override FPP0 Port filtering end address
override_FPPFilteringStart1	Uns32	Override FPP1 Port filtering start address
override_FPPFilteringEnd1	Uns32	Override FPP1 Port filtering end address
override_FPPFilteringStart2	Uns32	Override FPP2 Port filtering start address
override_FPPFilteringEnd2	Uns32	Override FPP2 Port filtering end address
override_FPPFilteringStart3	Uns32	Override FPP3 Port filtering start address
override_FPPFilteringEnd3	Uns32	Override FPP3 Port filtering end address

Table 8.1: Parameters that can be set in: MPCORE

8.1 Parameter values and limits

These are the formal parameter limits and actual parameter values

Name	Min	Max	Default	Actual
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(Others)				
variant				Cortex-R8MPx3
verbose			t	t
suppressCPSWarnings			f	f
showHiddenRegs			f	f
UAL			t	t
disableGICModel			f	f
enableGICv2_64kB_Page			f	f
enableVFPAAtReset			f	f
useInternalTCMs			f	f
enableSystemBus			f	f
enableSystemMonitorBus			f	f
compatibility			ISA	ISA
unpredictableR15			undefined	undefined
unpredictableModal			nop	nop
maxSIMDUnroll	1	16	2	2
override_debugMask	0	0xffffffff	0	0
thumbNoCond			f	f
enableHostAtomics			f	f
enableWFEHostAtomics			f	f
endian			none	none
override_numCPUs	0	32	3	3
override_affinityMask	0	0xffffffff	0	0
override_MPIDR_MT			f	f
override_MPIDR_Aff0	0	255	0	0
override_MPIDR_Aff1	0	255	0	0
override_MPIDR_Aff2	0	255	0	0
override_MPIDR_Aff3	0	255	0	0
override_fcsePresent			t	t
override_fpexcDexPresent			f	f
override_advSIMDPresent			f	f
override_vfpPresent			f	f
override_timerScaleFactor	1	0x1ff	2	2
override_GICD_NSACRPresent			f	f
override_GICD_PPISRPresent			f	f
override_GICD_SPISRPresent			f	f
override_GIC_PPIMask	0	0xffff	0	0
override_GICCDISABLE			f	f
allowMSRToCurrentBank			f	f
override_SCTLR_V			f	f
override_SCTLR_IE			f	f
override_SCTLR_EE			f	f
override_SCTLR_TE			f	f
override_SCTLR_NMFI			f	f
override_SCTLR_CP15BEN_Present			f	f

override_MIDR	0	0xffffffff	0x410fc183	0x410fc183
override_CTR	0	0xffffffff	0x8333c003	0x8333c003
override_MPUIR	0	0xffffffff	0xc00	0xc00
override_MPUIR_DRegion	0	0xffffffff	12	12
override_CLIDR	0	0xffffffff	0x9200003	0x9200003
override_ACTLR_SMP			f	f
override_TCMTR	0	0xffffffff	0x10001	0x10001
override_AIDR	0	0xffffffff	0	0
override_ITCMRR_E			f	f
override_ITCMRR_SIZE	0	0xffffffff	4	4
override_tcmsPresent			t	t
override_DTCMRR_SIZE	0	0xffffffff	4	4
override_CBAR	0	0xffffffff	0xae000000	0xae000000
override_PFR0	0	0xffffffff	0x131	0x131
override_PFR1	0	0xffffffff	1	1
override_DFR0	0	0xffffffff	0x10404	0x10404
override_AFR0	0	0xffffffff	0	0
override_MMFR0	0	0xffffffff	0x110130	0x110130
override_MMFR1	0	0xffffffff	0	0
override_MMFR2	0	0xffffffff	0x1200000	0x1200000
override_MMFR3	0	0xffffffff	0x2111	0x2111
override_ISAR0	0	0xffffffff	0x2101111	0x2101111
override_ISAR1	0	0xffffffff	0x13112111	0x13112111
override_ISAR2	0	0xffffffff	0x21232141	0x21232141
override_ISAR3	0	0xffffffff	0x1112131	0x1112131
override_ISAR4	0	0xffffffff	0x10142	0x10142
override_ISAR5	0	0xffffffff	0	0
override_PMCR	0	0xffffffff	0x41182000	0x41182000
override_PMCEID0	0	0xffffffffffffff	0	0
override_PMCEID1	0	0xffffffffffffff	0	0
override_SACTLR	0	0xffffffff	0	0
override_BuildOptions0	0	0xffffffff	0	0
override_BuildOptions1	0	0xffffffff	0	0
override_DBGDIDR	0	0xffffffff	0	0
override_FPSID	0	0xffffffff	0x41023180	0x41023180
override_MVFR0	0	0xffffffff	0x10110221	0x10110221
override_MVFR1	0	0xffffffff	0x1000011	0x1000011
override_FPEXC	0	0xffffffff	0	0
override_GICC_IIDR	0	0xffffffff	0x3901043b	0x3901043b
override_GICD_TYPER	0	0xffffffff	2	2
override_GICD_TYPER_ITLines	0	31	2	2
override_GICD_ICFGRN	0	0xffffffff	0	0
override_GICD_IIDR	0	0xffffffff	0x400043b	0x400043b
override_GICH_VTR	0	0xffffffff	0	0
override_ICCPMRBits	4	8	5	5

override_minICCBPR	0	7	2	2
override_ERG	3	11	3	3
override_CCSIDR_1I	0	0xffffffffffff	0x703fe019	0x703fe019
override_CCSIDR_1D	0	0xffffffffffff	0x703fe019	0x703fe019
override_CCSIDR_2I	0	0xffffffffffff	0	0
override_CCSIDR_2D	0	0xffffffffffff	0	0
override_CCSIDR_3I	0	0xffffffffffff	0	0
override_CCSIDR_3D	0	0xffffffffffff	0	0
override_CCSIDR_4I	0	0xffffffffffff	0	0
override_CCSIDR_4D	0	0xffffffffffff	0	0
override_CCSIDR_5I	0	0xffffffffffff	0	0
override_CCSIDR_5D	0	0xffffffffffff	0	0
override_CCSIDR_6I	0	0xffffffffffff	0	0
override_CCSIDR_6D	0	0xffffffffffff	0	0
override_CCSIDR_7I	0	0xffffffffffff	0	0
override_CCSIDR_7D	0	0xffffffffffff	0	0
override_STROffsetPC12			t	t
override_mpuV5ExtAP			f	f
override_ignoreBadCp15			f	f
override_SGIDisable			f	f
override_condUndefined			f	f
override_deviceStrongAligned			t	t
override_Control_V			f	f
override_MainId	0	0xffffffff	0x410fc183	0x410fc183
override_CacheType	0	0xffffffff	0x8333c003	0x8333c003
override_MPUType	0	0xffffffff	0xc00	0xc00
override_InstructionAttributes0	0	0xffffffff	0x2101111	0x2101111
override_InstructionAttributes1	0	0xffffffff	0x13112111	0x13112111
override_InstructionAttributes2	0	0xffffffff	0x21232141	0x21232141
override_InstructionAttributes3	0	0xffffffff	0x1112131	0x1112131
override_InstructionAttributes4	0	0xffffffff	0x10142	0x10142
override_InstructionAttributes5	0	0xffffffff	0	0
override_SCUControl_Address_filtering_enable			f	f
override_MasterFilteringStart	0	0xffffffff	0xfff00000	0xfff00000
override_MasterFilteringEnd	0	0xffffffff	0	0
override_PeripheralsFilteringStart	0	0xffffffff	0xfff00000	0xfff00000
override_PeripheralsFilteringEnd	0	0xffffffff	0	0
override_FPPFilteringStart0	0	0xffffffff	0xfff00000	0xfff00000
override_FPPFilteringEnd0	0	0xffffffff	0	0
override_FPPFilteringStart1	0	0xffffffff	0xfff00000	0xfff00000
override_FPPFilteringEnd1	0	0xffffffff	0	0
override_FPPFilteringStart2	0	0xffffffff	0xfff00000	0xfff00000
override_FPPFilteringEnd2	0	0xffffffff	0	0
override_FPPFilteringStart3	0	0xffffffff	0xfff00000	0xfff00000
override_FPPFilteringEnd3	0	0xffffffff	0	0

Table 8.2: Parameter values and limits

Name	Type	Description
enableVFPAAtReset	Boolean	Enable vector floating point (SIMD and VFP) instructions at reset. (Enables cp10/11 in CPACR and sets FPEXC.EN)
override_advSIMDPresent	Boolean	Specifies that Advanced SIMD extensions are present (if true)
override_vfpPresent	Boolean	Specifies that VFP extensions are present (if true)
override_SCTLR_V	Boolean	Override SCTLR.V with the passed value (enables high vectors; also configurable using VINITHI pin)
override_SCTLR_IE	Boolean	Override SCTLR.IE with the passed value (configures instruction endianness; also configurable using CFGIE pin)
override_SCTLR_EE	Boolean	Override SCTLR.EE with the passed value (configures exception data endianness; also configurable using CFGEE pin)
override_SCTLR_TE	Boolean	Override SCTLR.TE with the passed value (configures Thumb state for exception handling; also configurable using TEINIT pin)
override_SCTLR_NMFI	Boolean	Override SCTLR.NMFI with the passed value (configures NMFI state for exception handling; also configurable using CFGNMFI pin)
override_SCTLR_CP15BEN_Present	Boolean	Enable ARMv7 SCTLR.CP15BEN bit (CP15 barrier enable)
override_ACTLR_SMP	Boolean	Override ACTLR SMP enable bit
override_TCMTR	Uns32	Override TCMTR register
override_ITCMRR_E	Boolean	Override ITCMRR register enable bit
override_ITCMRR_SIZE	Uns32	Override ITCMRR register size bits
override_tcmsPresent	Boolean	Specifies that TCM is present (if true)
override_DTCMRR_SIZE	Uns32	Override DTCMRR register size bits

Table 8.3: Parameters that can be set in: CPU

8.2 Parameter values and limits

These are the formal parameter limits and actual parameter values

Name	Min	Max	Default	Actual
(Others)				
enableVFPAAtReset			f	f
override_advSIMDPresent			f	f
override_vfpPresent			f	f
override_SCTLR_V			f	f
override_SCTLR_IE			f	f
override_SCTLR_EE			f	f
override_SCTLR_TE			f	f
override_SCTLR_NMFI			f	f
override_SCTLR_CP15BEN_Present			f	f
override_ACTLR_SMP			f	f
override_TCMTR	0	0xffffffff	0x10001	0x10001

override_ITCMRR_E			f	f
override_ITCMRR_SIZE	0	0xffffffff	4	4
override_tcmsPresent			t	t
override_DTCMRR_SIZE	0	0xffffffff	4	4

Table 8.4: Parameter values and limits

Chapter 9

Execution Modes

Mode	Code
User	16
FIQ	17
IRQ	18
Supervisor	19
Abort	23
Undefined	27
System	31

Table 9.1: Modes implemented in: CPU

Chapter 10

Exceptions

Exception	Code
Reset	0
Undefined	1
SupervisorCall	2
PrefetchAbort	5
DataAbort	6
IRQ	8
FIQ	9

Table 10.1: Exceptions implemented in: CPU

Chapter 11

Hierarchy of the model

A CPU core may be configured to instance many processors of a Symmetrical Multi Processor (SMP). A CPU core may also have sub elements within a processor, for example hardware threading blocks.

OVP processor models can be written to include SMP blocks and to have many levels of hierarchy. Some OVP CPU models may have a fixed hierarchy, and some may be configured by settings in a configuration register. Please see the register definitions of this model.

This model documentation shows the settings and hierarchy of the default settings for this model variant.

11.1 Level 1: MPCORE

This level in the model hierarchy has 2 commands.

This level in the model hierarchy has no register groups.

This level in the model hierarchy has 3 children:

CPU0, CPU1 and CPU2.

11.2 Level 2: CPU

This level in the model hierarchy has 4 commands.

This level in the model hierarchy has 17 register groups:

Group name	Registers
Core	16
Control	3
User	7
FIQ	8
IRQ	3
Supervisor	3
Undefined	3
Abort	3
SIMD_VFP	16
SIMD_VFP_SYS	5
Coprocessor_32_bit	80

Integration_support	6
MPCore_SCR	18
MPCore_distributor	85
MPCore_processor_interface	8
MPCore_timer_and_watchdog	10
MPCore_global_timer	5

Table 11.1: Register groups

This level in the model hierarchy has no children.

Chapter 12

Model Commands

A Processor model can implement one or more **Model Commands** available to be invoked from the simulator command line, from the OP API or from the Imperas Multiprocessor Debugger.

12.1 Level 1: MPCORE

12.1.1 isync

specify instruction address range for synchronous execution

Argument	Type	Description
-addresshi	Uns64	end address of synchronous execution range
-addresslo	Uns64	start address of synchronous execution range

Table 12.1: isync command arguments

12.1.2 itrace

enable or disable instruction tracing

Argument	Type	Description
-access	String	show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-after	Uns64	apply after this many instructions
-enable	Boolean	enable instruction tracing
-full	Boolean	turn on all trace features
-instructioncount	Boolean	include the instruction number in each trace
-memory	String	(Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-mode	Boolean	show processor mode changes
-off	Boolean	disable instruction tracing
-on	Boolean	enable instruction tracing
-processorname	Boolean	Include processor name in all trace lines

-registerchange	Boolean	show registers changed by this instruction
-registers	Boolean	show registers after each trace

Table 12.2: itrace command arguments

12.2 Level 2: CPU

12.2.1 debugflags

show or modify the processor debug flags

Argument	Type	Description
-get	Boolean	print current processor flags value
-mask	Boolean	print valid debug flag bits
-set	Int32	new processor flags (only flags 0x000003e4 can be modified)

Table 12.3: debugflags command arguments

12.2.2 isync

specify instruction address range for synchronous execution

Argument	Type	Description
-addresshi	Uns64	end address of synchronous execution range
-addresslo	Uns64	start address of synchronous execution range

Table 12.4: isync command arguments

12.2.3 itrace

enable or disable instruction tracing

Argument	Type	Description
-access	String	show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-after	Uns64	apply after this many instructions
-enable	Boolean	enable instruction tracing
-full	Boolean	turn on all trace features
-instructioncount	Boolean	include the instruction number in each trace
-memory	String	(Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-mode	Boolean	show processor mode changes
-off	Boolean	disable instruction tracing
-on	Boolean	enable instruction tracing

-processorname	Boolean	Include processor name in all trace lines
-registerchange	Boolean	show registers changed by this instruction
-registers	Boolean	show registers after each trace

Table 12.5: itrace command arguments

12.2.4 listSysRegsAA32

12.2.4.1 Argument description

List all AArch32 system registers

Chapter 13

Registers

13.1 Level 1: MPCORE

No registers.

13.2 Level 2: CPU

13.2.1 Core

Registers at level:2, type:CPU group:Core

Name	Bits	Initial-Hex	RW	Description
r0	32	0	rw	
r1	32	0	rw	
r2	32	0	rw	
r3	32	0	rw	
r4	32	0	rw	
r5	32	0	rw	
r6	32	0	rw	
r7	32	0	rw	
r8	32	0	rw	
r9	32	0	rw	
r10	32	0	rw	
r11	32	0	rw	frame pointer
r12	32	0	rw	
sp	32	0	rw	stack pointer
lr	32	0	rw	
pc	32	0	rw	program counter

Table 13.1: Registers at level 2, type:CPU group:Core

13.2.2 Control

Registers at level:2, type:CPU group:Control

Name	Bits	Initial-Hex	RW	Description
fps	32	0	rw	archaic FPSCR view (for gdb)
cpsr	32	1d3	rw	
spsr	32	0	rw	

Table 13.2: Registers at level 2, type:CPU group:Control

13.2.3 User

Registers at level:2, type:CPU group:User

Name	Bits	Initial-Hex	RW	Description
r8_usr	32	0	rw	
r9_usr	32	0	rw	
r10_usr	32	0	rw	
r11_usr	32	0	rw	
r12_usr	32	0	rw	
sp_usr	32	0	rw	
lr_usr	32	0	rw	

Table 13.3: Registers at level 2, type:CPU group:User

13.2.4 FIQ

Registers at level:2, type:CPU group:FIQ

Name	Bits	Initial-Hex	RW	Description
r8_fiq	32	0	rw	
r9_fiq	32	0	rw	
r10_fiq	32	0	rw	
r11_fiq	32	0	rw	
r12_fiq	32	0	rw	
sp_fiq	32	0	rw	
lr_fiq	32	0	rw	
spsr_fiq	32	0	rw	

Table 13.4: Registers at level 2, type:CPU group:FIQ

13.2.5 IRQ

Registers at level:2, type:CPU group:IRQ

Name	Bits	Initial-Hex	RW	Description
sp_irq	32	0	rw	
lr_irq	32	0	rw	
spsr_irq	32	0	rw	

Table 13.5: Registers at level 2, type:CPU group:IRQ

13.2.6 Supervisor

Registers at level:2, type:CPU group:Supervisor

Name	Bits	Initial-Hex	RW	Description
sp_svc	32	0	rw	
lr_svc	32	0	rw	
spsr_svc	32	0	rw	

Table 13.6: Registers at level 2, type:CPU group:Supervisor

13.2.7 Undefined

Registers at level:2, type:CPU group:Undefined

Name	Bits	Initial-Hex	RW	Description
sp_undef	32	0	rw	
lr_undef	32	0	rw	
spsr_undef	32	0	rw	

Table 13.7: Registers at level 2, type:CPU group:Undefined

13.2.8 Abort

Registers at level:2, type:CPU group:Abort

Name	Bits	Initial-Hex	RW	Description
sp_abt	32	0	rw	
lr_abt	32	0	rw	
spsr_abt	32	0	rw	

Table 13.8: Registers at level 2, type:CPU group:Abort

13.2.9 SIMD_VFP

Registers at level:2, type:CPU group:SIMD_VFP

Name	Bits	Initial-Hex	RW	Description
d0	64	0	rw	
d1	64	0	rw	
d2	64	0	rw	
d3	64	0	rw	
d4	64	0	rw	
d5	64	0	rw	
d6	64	0	rw	
d7	64	0	rw	
d8	64	0	rw	
d9	64	0	rw	
d10	64	0	rw	
d11	64	0	rw	
d12	64	0	rw	
d13	64	0	rw	
d14	64	0	rw	
d15	64	0	rw	

Table 13.9: Registers at level 2, type:CPU group:SIMD_VFP

13.2.10 SIMD_VFP_SYS

Registers at level:2, type:CPU group:SIMD_VFP_SYS

Name	Bits	Initial-Hex	RW	Description
FPSID	32	41023180	r-	floating-point system ID
FPSCR	32	0	rw	floating-point status/control
FPEXC	32	0	rw	floating-point exception
MVFR0	32	10110221	r-	Media/VFP feature 0

MVFR1	32	1000011	r-	Media/VFP feature 1
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Table 13.10: Registers at level 2, type:CPU group:SIMD_VFP_SYS

13.2.11 Coprocessor_32_bit

Registers at level:2, type:CPU group:Coprocessor_32_bit

Name	Bits	Initial-Hex	RW	Description
ACTLR	32	0	rw	Auxiliary Control
ADFSR	32	0	rw	Auxiliary Data Fault Status
AIFSR	32	0	rw	Auxiliary Instruction Fault Status
BPIALL	32	-	-w	Branch Predictor Invalidate All
BPIALLIS	32	-	-w	Branch Predictor Invalidate All (IS)
BPIMVA	32	-	-w	Branch Predictor Invalidate by VA
CBAR	32	ae000000	rw	Configuration Base Address
CCSIDR	32	703fe019	r-	Cache Size ID
CLIDR	32	9200003	r-	Cache Level ID
CONTEXTIDR	32	0	rw	Context ID
CP15DMB	32	-	-w	CP15 Data Memory Barrier
CP15DSB	32	-	-w	CP15 Data Synchronization Barrier
CP15ISB	32	-	-w	CP15 Instruction Synchronization Barrier
CP15NOP	32	-	-w	CP15 NOP
CPACR	32	c0000000	rw	Coprocessor Access Control
CSSELR	32	0	rw	Cache Size Selection
CTR	32	8333c003	r-	Cache Type
DBGDIDR	32	0	r-	Debug ID
DCCIMVAC	32	-	-w	Data Cache Line Clean and Invalidate by VA to PoC
DCCISW	32	-	-w	Data Cache Line Clean and Invalidate by Set/Way
DCCMVAC	32	-	-w	Data Cache Line Clean by VA to PoC
DCCMVAU	32	-	-w	Data Cache Line Clean by VA to PoU
DCCSW	32	-	-w	Data Cache Line Clean by Set/Way
DCIMVAC	32	-	-w	Data Cache Line Invalidate by VA to PoC
DCISW	32	-	-w	Data Cache Line Invalidate by Set/Way
DFAR	32	0	rw	Data Fault Address
DFSR	32	0	rw	Data Fault Status
DRACR	32	0	rw	Data Region Access Control
DRBAR	32	0	rw	Data Region Base Address
DRSR	32	0	rw	Data Region Size/Enable
DTCMRR	32	10	rw	DTCM Region
FCSEIDR	32	0	rw	FCSE Process ID
ICIALLU	32	-	-w	Instruction Cache Invalidate All
ICIALLUIS	32	-	-w	Instruction Cache Invalidate All (IS)
ICIMVAU	32	-	-w	Instruction Cache Invalidate by VA
ID_AFR0	32	0	r-	Auxiliary Feature 0
ID_DFR0	32	10404	r-	Debug Feature 0
ID_ISAR0	32	2101111	r-	Instruction Set Attribute 0
ID_ISAR1	32	13112111	r-	Instruction Set Attribute 1
ID_ISAR2	32	21232141	r-	Instruction Set Attribute 2
ID_ISAR3	32	1112131	r-	Instruction Set Attribute 3
ID_ISAR4	32	10142	r-	Instruction Set Attribute 4
ID_ISAR5	32	0	r-	Instruction Set Attribute 5
ID_MMFR0	32	110130	r-	Memory Model Feature 0
ID_MMFR1	32	0	r-	Memory Model Feature 1
ID_MMFR2	32	1200000	r-	Memory Model Feature 2

ID_MMFR3	32	2111	r-	Memory Model Feature 3
ID_PFR0	32	131	r-	Processor Feature 0
ID_PFR1	32	1	r-	Processor Feature 1
IFAR	32	0	rw	Instruction Fault Address
IFSR	32	0	rw	Instruction Fault Status
ISR	32	0	r-	Interrupt Status
ITCMRR	32	10	rw	ITCM Region
JIDR	32	0	rw	Jazelle ID
JMCR	32	0	rw	Jazelle Main Configuration
JOSCR	32	0	rw	Jazelle OS Control
L2CTLR	32	2000000	rw	L2 Control
L2ECTLR	32	0	rw	L2 Extended Control
MIDR	32	410fc183	r-	Main ID
MPIDR	32	80000000	r-	Multiprocessor Affinity
MPUIR	32	c00	r-	MPU Type
PAR	32	0	rw	Physical Address
PMCCNTR	32	0	rw	Performance Monitors Cycle Count
PMCNTENCLR	32	0	rw	Performance Monitors Count Enable Clear
PMCNTENSET	32	0	rw	Performance Monitors Count Enable Set
PMCR	32	41182000	rw	Performance Monitors Control
PMINTENCLR	32	0	rw	Performance Monitors Interrupt Enable Clear
PMINTENSET	32	0	rw	Performance Monitors Interrupt Enable Set
PMOVSr	32	0	rw	Performance Monitors Overflow Flag Status
PMSELR	32	0	rw	Performance Monitors Event Counter Selection
PMSWINC	32	-	-w	Performance Monitors Software Increment
PMUSERENR	32	0	rw	Performance Monitors User Enable
PMXEVCNTR	32	0	rw	Performance Monitors Selected Event Count
PMXEVTPER	32	0	rw	Performance Monitors Selected Event Type
RGNR	32	0	rw	Region Number
SCTLR	32	0	rw	System Control
TCMTR	32	10001	r-	TCM Type
TPIDRPRW	32	0	rw	PL0 Read/Write Software Thread ID
TPIDRURO	32	0	rw	PL0 Read-Only Software Thread ID
TPIDRURW	32	0	rw	PL1 Software Thread ID

Table 13.11: Registers at level 2, type:CPU group:Coprocessor_32_bit

13.2.12 Integration_support

Registers at level:2, type:CPU group:Integration_support

Name	Bits	Initial-Hex	RW	Description
transactPL	32	1	r-	privilege level of current memory transaction
transactAT	32	0	r-	current memory transaction type: PA=1, VA=0
HaltReason	8	0	r-	bit field indicating halt reason
atomicType	8	0	r-	current atomic instruction type (1:atomic, 2:exclusive)
WakeWFINow	8	0	-w	wake up core from WFI state
ambaResponse	64	0	rw	external AMBA-PV response

Table 13.12: Registers at level 2, type:CPU group:Integration_support

13.2.13 MPCore_SCR

Registers at level:2, type:CPU group:MPCore_SCR

Name	Bits	Initial-Hex	RW	Description
FPPFilteringEnd0	32	0	rw	FPP0 Filtering End Address
FPPFilteringEnd1	32	0	rw	FPP1 Filtering End Address
FPPFilteringEnd2	32	0	rw	FPP2 Filtering End Address
FPPFilteringEnd3	32	0	rw	FPP3 Filtering End Address
FPPFilteringStart0	32	fff00000	rw	FPP0 Filtering Start Address
FPPFilteringStart1	32	fff00000	rw	FPP1 Filtering Start Address
FPPFilteringStart2	32	fff00000	rw	FPP2 Filtering Start Address
FPPFilteringStart3	32	fff00000	rw	FPP3 Filtering Start Address
MasterFilteringEnd	32	0	rw	Master Filtering End
MasterFilteringStart	32	fff00000	rw	Master Filtering Start
PeripheralsFilteringEnd	32	0	rw	Peripheral Filtering End
PeripheralsFilteringStart	32	fff00000	rw	Peripheral Filtering Start
SCUCPUPowerStatus	32	3000000	rw	SCU CPU Power Status
SCUConfiguration	32	fff02	r-	SCU Configuration
SCUControl	32	0	rw	SCU Control
SCUInvalidateSecure	32	0	-w	SCU Invalidate Secure
SCUNSAC	32	0	rw	SCU Non-secure Access Control
SCUSAC	32	f	rw	SCU Secure Access Control

Table 13.13: Registers at level 2, type:CPU group:MPCore.SCR

13.2.14 MPCore_distributor

Registers at level:2, type:CPU group:MPCore_distributor

Name	Bits	Initial-Hex	RW	Description
ICCIDR0	32	d	r-	Component ID 0
ICCIDR1	32	90	r-	Component ID 1
ICCIDR2	32	5	r-	Component ID 2
ICCIDR3	32	b1	r-	Component ID 3
ICDABR0	32	0	rw	Interrupt Set-Active 0
ICDABR1	32	0	rw	Interrupt Set-Active 1
ICDABR2	32	0	rw	Interrupt Set-Active 2
ICDDCR	32	0	rw	Distributor Control
ICDICER0	32	fff	rw	Interrupt Clear-Enable 0
ICDICER1	32	0	rw	Interrupt Clear-Enable 1
ICDICER2	32	0	rw	Interrupt Clear-Enable 2
ICDICFR0	32	aaaaaaaa	rw	Interrupt Configuration 0
ICDICFR1	32	7dc00000	rw	Interrupt Configuration 1
ICDICFR2	32	0	rw	Interrupt Configuration 2
ICDICFR3	32	0	rw	Interrupt Configuration 3
ICDICFR4	32	0	rw	Interrupt Configuration 4
ICDICFR5	32	0	rw	Interrupt Configuration 5
ICDICPR0	32	0	rw	Interrupt Clear-Pending 0
ICDICPR1	32	0	rw	Interrupt Clear-Pending 1
ICDICPR2	32	0	rw	Interrupt Clear-Pending 2
ICDICTR	32	42	r-	Interrupt Controller Type
ICDIIDR	32	400043b	r-	Distributor Implementor ID
ICDIPR0	32	0	rw	Interrupt Priority 0
ICDIPR1	32	0	rw	Interrupt Priority 1
ICDIPR2	32	0	rw	Interrupt Priority 2
ICDIPR3	32	0	rw	Interrupt Priority 3
ICDIPR4	32	0	rw	Interrupt Priority 4
ICDIPR5	32	0	rw	Interrupt Priority 5

ICDIPR6	32	0	rw	Interrupt Priority 6
ICDIPR7	32	0	rw	Interrupt Priority 7
ICDIPR8	32	0	rw	Interrupt Priority 8
ICDIPR9	32	0	rw	Interrupt Priority 9
ICDIPR10	32	0	rw	Interrupt Priority 10
ICDIPR11	32	0	rw	Interrupt Priority 11
ICDIPR12	32	0	rw	Interrupt Priority 12
ICDIPR13	32	0	rw	Interrupt Priority 13
ICDIPR14	32	0	rw	Interrupt Priority 14
ICDIPR15	32	0	rw	Interrupt Priority 15
ICDIPR16	32	0	rw	Interrupt Priority 16
ICDIPR17	32	0	rw	Interrupt Priority 17
ICDIPR18	32	0	rw	Interrupt Priority 18
ICDIPR19	32	0	rw	Interrupt Priority 19
ICDIPR20	32	0	rw	Interrupt Priority 20
ICDIPR21	32	0	rw	Interrupt Priority 21
ICDIPR22	32	0	rw	Interrupt Priority 22
ICDIPR23	32	0	rw	Interrupt Priority 23
ICDIPTR0	32	1010101	rw	Interrupt Processor Targets 0
ICDIPTR1	32	1010101	rw	Interrupt Processor Targets 1
ICDIPTR2	32	1010101	rw	Interrupt Processor Targets 2
ICDIPTR3	32	1010101	rw	Interrupt Processor Targets 3
ICDIPTR4	32	0	rw	Interrupt Processor Targets 4
ICDIPTR5	32	0	rw	Interrupt Processor Targets 5
ICDIPTR6	32	1000000	rw	Interrupt Processor Targets 6
ICDIPTR7	32	1010101	rw	Interrupt Processor Targets 7
ICDIPTR8	32	0	rw	Interrupt Processor Targets 8
ICDIPTR9	32	0	rw	Interrupt Processor Targets 9
ICDIPTR10	32	0	rw	Interrupt Processor Targets 10
ICDIPTR11	32	0	rw	Interrupt Processor Targets 11
ICDIPTR12	32	0	rw	Interrupt Processor Targets 12
ICDIPTR13	32	0	rw	Interrupt Processor Targets 13
ICDIPTR14	32	0	rw	Interrupt Processor Targets 14
ICDIPTR15	32	0	rw	Interrupt Processor Targets 15
ICDIPTR16	32	0	rw	Interrupt Processor Targets 16
ICDIPTR17	32	0	rw	Interrupt Processor Targets 17
ICDIPTR18	32	0	rw	Interrupt Processor Targets 18
ICDIPTR19	32	0	rw	Interrupt Processor Targets 19
ICDIPTR20	32	0	rw	Interrupt Processor Targets 20
ICDIPTR21	32	0	rw	Interrupt Processor Targets 21
ICDIPTR22	32	0	rw	Interrupt Processor Targets 22
ICDIPTR23	32	0	rw	Interrupt Processor Targets 23
ICDISER0	32	fff	rw	Interrupt Set-Enable 0
ICDISER1	32	0	rw	Interrupt Set-Enable 1
ICDISER2	32	0	rw	Interrupt Set-Enable 2
ICDISPR0	32	0	rw	Interrupt Set-Pending 0
ICDISPR1	32	0	rw	Interrupt Set-Pending 1
ICDISPR2	32	0	rw	Interrupt Set-Pending 2
ICDSGIR	32	0	-w	Software-Generated Interrupt
ICPIDR0	32	18	r-	Peripheral ID 0
ICPIDR1	32	bc	r-	Peripheral ID 1
ICPIDR2	32	b	r-	Peripheral ID 2
ICPIDR3	32	0	r-	Peripheral ID 3
ICPIDR4	32	4	r-	Peripheral ID 4
ICPIDR5	32	0	r-	Peripheral ID 5
ICPIDR6	32	0	r-	Peripheral ID 6

ICPIDR7	32	0	r-	Peripheral ID 7
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Table 13.14: Registers at level 2, type:CPU group:MPCore_distributor

13.2.15 MPCore_processor_interface

Registers at level:2, type:CPU group:MPCore_processor_interface

Name	Bits	Initial-Hex	RW	Description
ICCBPR	32	0	rw	Binary Point
ICCEOIR	32	0	-w	End of Interrupt
ICCHPIR	32	3ff	r-	Highest Priority Pending Interrupt
ICCIAR	32	3ff	r-	Interrupt Acknowledge
ICCICR	32	0	rw	CPU Interface Control
ICCHDR	32	3901043b	r-	CPU Interface ID
ICCPMR	32	0	rw	Interrupt Priority Mask
ICCRPR	32	ff	r-	Running Priority

Table 13.15: Registers at level 2, type:CPU group:MPCore_processor_interface

13.2.16 MPCore_timer_and_watchdog

Registers at level:2, type:CPU group:MPCore_timer_and_watchdog

Name	Bits	Initial-Hex	RW	Description
PTControl	32	0	rw	Private Timer Control
PTCounter	32	0	rw	Private Timer Counter
PTInterruptStatus	32	0	rw	Private Timer Interrupt Status
PTLoad	32	0	rw	Private Timer Load
WTControl	32	0	rw	Watchdog Timer Control
WTCounter	32	0	rw	Watchdog Timer Counter
WTDisable	32	0	-w	Watchdog Timer Disable
WTInterruptStatus	32	0	rw	Watchdog Timer Interrupt Status
WTLload	32	0	rw	Watchdog Timer Load
WTRResetStatus	32	0	rw	Watchdog Timer Reset Status

Table 13.16: Registers at level 2, type:CPU group:MPCore_timer_and_watchdog

13.2.17 MPCore_global_timer

Registers at level:2, type:CPU group:MPCore_global_timer

Name	Bits	Initial-Hex	RW	Description
GTAutoIncrement	32	0	rw	Global Timer Auto-Increment
GTComparator	64	0	rw	Global Timer Comparator
GTControl	32	0	rw	Global Timer Control
GTCounter	64	0	rw	Global Timer Counter
GTInterruptStatus	32	0	rw	Global Timer Interrupt Status

Table 13.17: Registers at level 2, type:CPU group:MPCore_global_timer